Indian Institute of Technology Kharagpur
Class Test III 2024-25

Date of Examination: 18 Mar, 2025 **Duration:** 30 Minutes
Subject No.: CS60050 **Subject:** Machine Learning
Department/Center/School: Computer Science **Credits:** 3 **Full marks:** 30

Name: ______________________________

Roll Number: ____________________

Question:	1	2	3	Total
Points:	10	10	10	30
Score:				

Write the answers in the space provided after the questions only.

1. (10 points) Consider the following dataset with an initial clustering $D = \{C_1 = [(0,2),(1,2)], C_2 = [(2,1),(2,0)]\}$. Answer the following questions:

   (a) Write the maximum-likelihood estimates for the parameters of the two clusters (μ_1, Σ_1), (μ_2, Σ_2)

   Solution:
   $\mu_1 = (0.5, 2)$, $\Sigma_1 = \begin{pmatrix} 0.25 & 0 \\ 0 & 0 \end{pmatrix}$

   $\mu_2 = (2, 0.5)$, $\Sigma_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0.25 \end{pmatrix}$

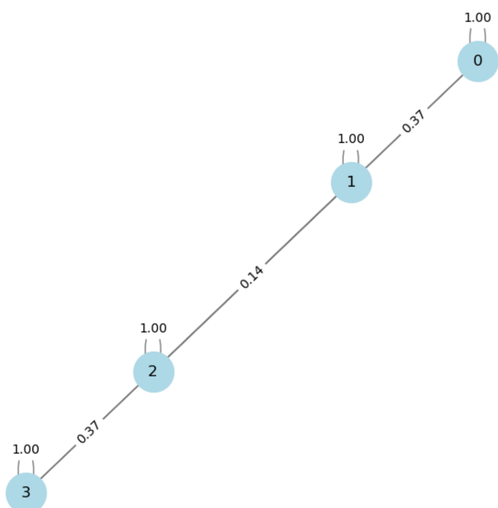
   (b) Write the formula for computing $P(z_k = 1|x)$ and compute the probability of $(0,2)$ belonging to cluster C_2, considering the prior probability of both clusters to be $1/2$.

Solution:

$P(z_k = 1|x) = \frac{P(z_k=1)P(x|z_k=1)}{\sum_{k\in\{1,2\}} P(z_k=1)P(x|z_k=1)}$

$P(z_2 = 1|(0,2)) = \frac{0.5*\mathcal{N}((0,2)|\mu_2,\Sigma_2)}{0.5*(\mathcal{N}((0,2)|\mu_1,\Sigma_1)+\mathcal{N}((0,2)|\mu_2,\Sigma_2))}$

2. (10 points) (a) Consider the following adjacency matrix for a given dataset. Calculate the un-normalized laplacian matrix and the weighted degree for each node.



Solution:

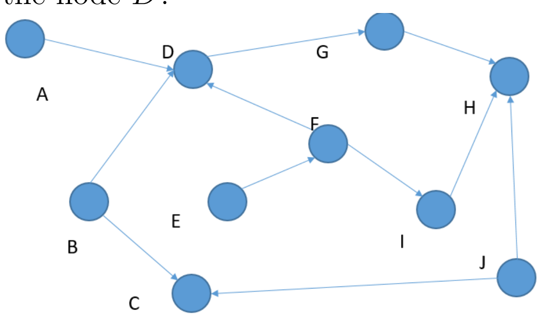
$$\text{Laplacian} = \begin{bmatrix} 0.37 & -0.37 & 0 & 0 \\ -0.37 & 0.51 & -0.14 & 0 \\ 0 & -0.14 & 0.51 & -0.37 \\ 0 & 0 & -0.37 & 0.37 \end{bmatrix}$$

(b) What is the lowest eigenvalue and the corresponding eigenvector for the unnormalized Laplacian.

Solution:

The lowest eigenvalue is 0 and the corresponding eigenvector is one with all elements as the same value. The value could be anything for unnormalized eigenvectors. For normalized eigenvectors it is $1/\sqrt{d}$ or $-1/\sqrt{d}$.

3. (10 points) (a) Consider the following bayesian network. What is the markov blanket around the node D?



Solution:

A, B, F, G

Markov blanket is the set of nodes that separate a given node from all the other nodes. This can be derived using the just the rules of D-separation.

(b) Write the expression for the probability $P(D|E)$ involving the minimal set of other random variables.

Solution:

It is sufficient to consider only the markov blanket around D and E since all the other nodes are conditionally independent.

$$\begin{aligned} P(D|E) &= \sum_{A,B,F,G} P(A, B, F, D, G|E) \\ &= \sum_{A,B,F,G} P(A)P(B)P(F|E)P(D|E, F))P(G|D, E, F) \\ &= \sum_{F,G} P(F|E)P(D|F)P(G|D) \end{aligned}$$

Rough space